Kernel ETW is the best ETW

This research focuses on the importance of native audit logs in secure by design software, emphasizing the need for kernel level ETW logging over user-mode hooks to enhance anti-tamper protections.

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Preamble

A critical feature of secure by design software is the generation of audit logs when privileged operations are performed. These native audit logs can be used for detection and response, and for forensic analysis. Some of these logs are quite easy to enable, but others are harder to enable and to alter then, for.

Most Windows components generate logs using **Event Tracing for Windows** (ETW). These events expose some of Windows's inner workings, and there are scenarios where independent security tools may want to consume these events, for example, to detect malicious activity. Although, not all ETW providers are created equal.

For example, an attacker may want to disable ETW logging, for instance, to **evade** detection. Some ETW providers are harder to alter than others, for example, disabling the kernel-mode ETW providers is harder than disabling a **user-mode** ETW provider. Some ETW providers are harder to alter than others, for example, disabling the kernel-mode ETW providers is harder than disabling a user-mode